SHEKH EHSANUR RAHMAN

Senior Mobile & Full-Stack AI Engineer | Flutter · Native iOS · Android · macOS

Flutter Architecture · Custom Plugins & Native Integration · Offline-First Sync · Payments & Subscriptions · Claude · FastAPI · Postgres

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Rajshahi, Bangladesh (UTC+6) · Remote · **No relocation or visa sponsorship required** · Full-time or contract · Works with any time zone; flexible core overlap hours

PROFILE

Senior Mobile & Full-Stack AI Engineer with 7 years of Flutter inside 11 years of shipping mobile apps, and the native depth to go below the framework: platform channels and plugins in Swift, Kotlin and Java, home-screen widgets, and a native macOS app of my own. 9 Flutter packages published on pub.dev. I take on the problems that block a product: offline data that must not be lost, payments that must reconcile across Apple and Stripe, platform edge cases, and slow components that need a native rewrite. Highlights: a subscription app stalled for ~2 years went live within ~1 month of my payment redesign; a native scanner reads codes in ~0.5 s; releases cut from ~1 hour to 10–20 minutes. I build with AI coding agents and review every change myself. Founder of two products built solo: Hourwise, a native macOS app with on-device machine learning (Mac App Store), and Repha, a B2B ordering network for pharmacies and medical representatives in Bangladesh (Google Play).

AREAS OF EXPERTISE

Flutter & Dart (7 yrs) · Flutter Architecture (Clean Architecture, BLoC, Riverpod) · Flutter Plugin & Package Development · Platform Channels & Native Integration (Swift, Kotlin, Java) · Native iOS, Android & macOS · Offline-First & Data Synchronization · In-App Purchases & Subscriptions · Payment Integration · Flutter Testing (unit, widget, integration) · Crash Reporting & Monitoring · CI/CD & Release Automation · React, Node.js & Next.js · Laravel · PostgreSQL / MongoDB / Redis · App Store & Google Play Release · On-Device Machine Learning · AI-Assisted Development · Technical Leadership

PROFESSIONAL EXPERIENCE

Founder & Full-Stack AI Engineer — Hourwise Labs

09/2026 – Present · Remote · Self-employed · Mac App Store · ehsanur.com/hourwise · Google Play · ehsanur.com/repha

- Designed and built, alone, a native macOS menu-bar app that records billable work passively for solo lawyers and turns it into invoices; approved and live on the Mac App Store.
- Assigned work to client matters with a rule engine plus an on-device naive-Bayes classifier that learns from user corrections — chosen over a cloud model so client work history never leaves the Mac.
- SwiftUI + AppKit (~17k lines of Swift); local-first SQLite/GRDB with versioned migrations; PDFKit invoices with LEDES 1998B, QuickBooks and Xero exports; StoreKit 2. Built with AI coding agents, with architecture and review by me. In progress: AI billing narratives with Apple Foundation Models.
- Designed and built, alone, **Repha** — a B2B ordering network that connects Bangladeshi pharmacies with pharma companies' medical representatives; live on Google Play (10/2026), pilot in Rajshahi.
- Flutter app for both sides: 73,000+ medicines searchable offline (SQLite FTS5); one order list split automatically by company and routed to the right rep; on-device OCR turns a photo of a printed list into order lines; an offline outbox sends each order exactly once; Bangla-first UI.
- Laravel + MySQL API and a Filament admin panel where every limit and feature can be switched with full change history; push notifications; runs at zero hosting cost on shared hosting.

Senior Mobile Developer (Level III) — Developer eXperience Hub (Devxhub)

09/2023 – 09/2026 · Remote — Rajshahi, Bangladesh

- Team lead and mobile lead for client products in FinTech, subscriptions, marketplaces and consumer apps; owned architecture, state management and releases in Flutter with native Swift and Kotlin modules. Key projects below.
- **Offline-first sync** for expense and FinTech apps serving 50,000+ users: local-first writes with one deterministic conflict-resolution rule, so the same conflicting edits resolve the same way on every device; claims made without signal sync when the connection returns, and recurring sync issues fell.
- **Release pipeline:** Fastlane + GitHub Actions cut each release from ~1 hour to 10–20 minutes and removed the dependency on one developer's machine across time zones.
- **AI-assisted delivery:** introduced Claude Code to the team, with code review and tests required on every merge.

- **Production monitoring:** Sentry, Firebase Crashlytics and UXCam to find and fix production errors; Firebase Remote Config to force updates after major releases.

Senior Mobile Application Developer — Prabartan Information Technology

06/2019 – 08/2023 · Remote — Rajshahi, Bangladesh

- Led a 6-developer team building offline-first, multi-language learning apps for students on weak internet (project below).
- Modernised the stack (Flutter 3, Jetpack Compose, SwiftUI) and set code review standards for the team.

Android Application Developer — Blue Hills Information Technology

07/2015 – 05/2019 · On-site — Rajshahi, Bangladesh · Part-time alongside university through 04/2017

- Co-built consumer Android apps in Java and Kotlin that reached 500,000+ downloads and a top-10 Education ranking on Google Play; profiled app launch performance and introduced code review practices.

Guest Trainer, Mobile App Development (Contract) — NACTAR

2016 – 2024 · 2–3 sessions a year · Bogura, Bangladesh

- Designed and delivered hands-on Android and Flutter training to 200+ developers.

SELECTED PROJECTS

Repha — Pharmacy–Rep Ordering Network · *Flutter · Dart · SQLite FTS5 · ML Kit · Laravel · MySQL · Hourwise Labs* · Google Play · ehsanur.com/repha

- **Problem:** a Bangladeshi pharmacy buys from dozens of pharma companies, each through its own medical rep; orders live on paper lists and phone calls, get lost or arrive partly, and new shops don't know who the rep for their area is.
- **Decision:** one list instead of many calls — the app matches each line against a 73,000+ medicine catalog stored on the phone, splits the list by company and sends each part to that company's connected rep; built offline-first so orders are never lost on weak networks, and on shared hosting so the product costs nothing to run.
- **Result:** live on Google Play since October 2026, built solo end to end; pilot running in Rajshahi.

Native Receipt Scanner — Expense-Reimbursement FinTech · *Flutter · Swift · Java · WebSockets · Devxhub* · pub.dev package

- **Problem:** employees were reimbursed by scanning receipt QR codes and barcodes; the generic Flutter scanner was too slow and couldn't be customized, and verification showed only an endless spinner.
- **Decision:** rewrote scanning natively (Swift on iOS, Java on Android) behind a Flutter interface instead of tuning the plugin, for direct control over camera handling and scan performance; streamed verification progress to a live progress bar over WebSockets; let claims save offline and sync later.
- **Result:** ~0.5 s per code in testing; a successful reimbursement completes ~5–10 s after the scan. Extracted as an open-source package. The company has since closed, so the app listing is gone.

Cross-Platform Payment Entitlements — Subscription App · *Flutter · StoreKit 2 · Apple In-App Purchase · Stripe · backend · Devxhub* · App Store · Google Play

- **Problem:** Android paid through Stripe and iOS through Apple In-App Purchase, and the two disagreed: duplicate purchases, upgrades billed in full, access on the wrong account. The app had been in development for ~2 years.
- **Decision:** as team lead, with the backend developer, made one account-level entitlement on the backend the single source of truth for both payment systems — because Apple ties purchases to the Apple ID while the business sells to app accounts; added duplicate-purchase checks before accepting a payment and reconciled existing records.
- **Result:** App Store approval and launch ~1 month after stabilization began.

High-Latitude Prayer-Time Engine — Prayer-Times App · *Dart · Flutter · WidgetKit · Devxhub* · App Store · Google Play · pub.dev package

- **Problem:** users in far-northern cities saw prayer times that didn't match their real local schedule; earlier fixes by other developers had failed.
- **Decision:** traced the error to the sunrise/sunset calculation at high latitudes and replaced the existing calculation setup with our own offline engine, rather than patching individual locations; added offline location and time-zone data for 30+ countries with manual location selection.
- **Result:** correct times verified against a real local schedule, online or fully offline. Extracted as an open-source package.

Multi-Service Super App — Marketplace Platform · *Flutter · Dart · maps · live tracking · payments · Devxhub* · App Store · Google Play

- **Problem:** service providers and customers used separate apps, which made the product harder to use and grow as services were added.
- **Decision:** as team lead and sole mobile developer, merged them into one app that hosts each service, rather than adding more separate apps.
- **Result:** one app on both stores for ride-hailing with live tracking, on-demand services, food and grocery delivery, virtual consultations and listings.

Offline-First Learning Apps — EdTech · *Flutter · Firebase · SQLite · multi-language · Prabartan* · Google Play

- **Problem:** students on weak or no internet lost lessons and progress whenever the network dropped.
- **Decision:** made the device the source of truth — lessons and progress stored locally first — with a Firebase/SQLite sync layer and deterministic conflict resolution, instead of an online-only design.
- **Result:** two of the apps each passed 100,000 downloads on Google Play.

Also shipped (iOS and Android): e-commerce marketplace (App Store · Google Play), secure video-course platform with recording protection (App Store · Google Play), notes and task planner with cloud sync (App Store · Google Play), eSIM data-plan marketplace (Google Play).

TECHNICAL SKILLS

Languages: Dart, Swift, Kotlin, Java

Flutter: Flutter and Dart (7 yrs); custom plugins and platform channels to Swift, Kotlin and Java; CustomPainter and gesture handling; home-screen widgets sharing data with Flutter (WidgetKit, App Groups); multi-flavor builds; migration to Flutter 3; publishing and maintaining pub.dev packages

Native: Swift, SwiftUI, AppKit, Kotlin, Java, Jetpack Compose — iOS, Android, macOS; Xcode, Android Studio

Architecture & state: Clean Architecture, MVVM, dependency injection, repository pattern, BLoC, Cubit, Riverpod

Data & sync: offline-first and local-first architecture, conflict resolution, CRDT, Firebase (Firestore, Auth, Cloud Functions, Cloud Messaging, Remote Config), Supabase, SQLite (Drift, GRDB), SQLite FTS5 full-text search, REST APIs (incl. Laravel + MySQL), WebSockets, OAuth2/JWT

Payments: StoreKit 2, Apple In-App Purchase, Stripe, subscription and entitlement modelling

Quality & release: unit, widget and integration tests, code review, Sentry, Firebase Crashlytics, UXCam, Fastlane, GitHub Actions, code signing, App Store Connect, Google Play Console

AI: AI-assisted development with Claude Code; on-device machine learning (naive Bayes, live in production); Apple Foundation Models (in progress); computer vision (homography); on-device OCR (ML Kit)

Device & security: camera and QR/barcode scanning, maps and live location, biometric authentication, encrypted local storage, accessibility (WCAG)

OPEN SOURCE — 9 FLUTTER PACKAGES ON PUB.DEV (VERIFIED PUBLISHER)

- **From client work:** flutter_scanner_devxhub (native receipt scanner) and flutter_prayer_time_calculator (high-latitude prayer-time engine), both published under Devxhub.
- **My own:** flutter_crdt_sync_kit (CRDT layer that merges offline edits from several devices; Supabase/REST adapters), local_voice (on-device voice commands), flutter_a11y_lens (live WCAG auditing), plus flutter_fabric, flutter_liquid_glass_widgets (fork, original author credited), flutter_zoom_image and flutter_scroll_date_picker.

SIDE PROJECTS

- **OMR Scanner** — offline exam answer-sheet grading with registration-mark alignment and homography instead of OCR · github.com/sujit70777/omr_scanner_flutter
- **ar_measure** — Flutter plugin wrapping ARKit and ARCore for cross-platform AR distance and area measurement · github.com/sujit70777/ar_measure
- **Jatri** — transit journey planner and ticketing on real metro and bus route data · GitHub

EDUCATION & LANGUAGES

B.Sc. in Computer Science & Engineering — Varendra University, Rajshahi, Bangladesh · 2013 – 2017
Bangla — native · English — professional working proficiency